

NatureProtector - Current-State Report Supplement

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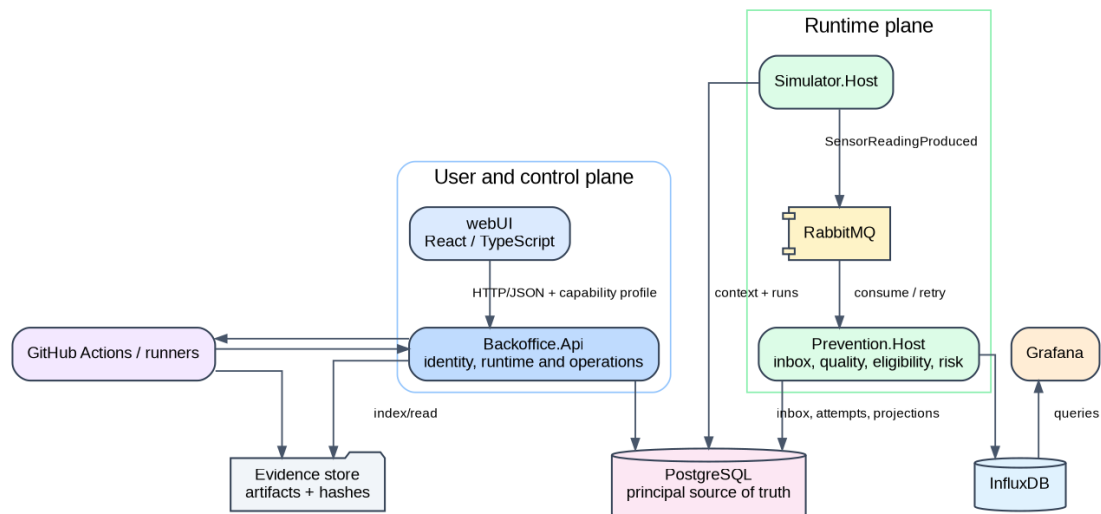
Purpose and authority

This supplement records the factual delta between the report state labelled **R13** and the repository snapshot that includes the Unified Operations Control Plane. It is an integration aid, not a replacement for the 379-page report and not evidence that staging or production were completed.

The supplied report presents the project as an experimental auditable platform and already distinguishes implementation, static verification, current result, historical execution, blocked attempt, reproduction and validation. The current repository extends that same evidence discipline to engineering operations.

Delta summary

The repository now contains server-side capability policies; the roles **QA**, **Operations** and **ReleaseApprover**; Mission Control, Quality Runs, Evidence Explorer, Deployments, Cloud Resources, Approvals and User/Role Administration; a closed operation catalog; operation records, confirmation and approval; GitHub workflow wrappers; and authenticated callback/artifact handling.



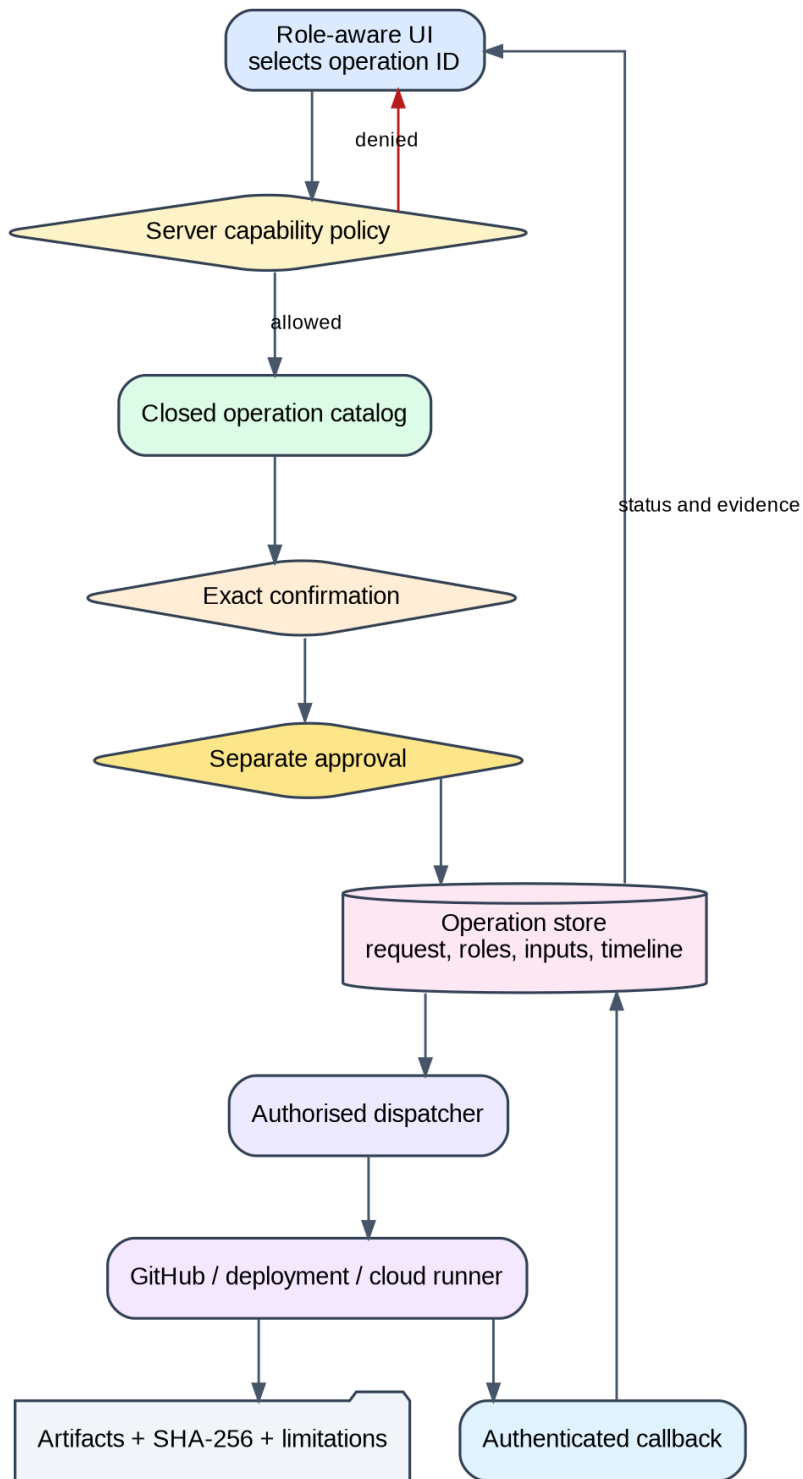
Current container architecture

Architecture integration

The report architecture should now distinguish three planes:

1. runtime/data processing;
2. user and control APIs;
3. engineering operations and evidence.

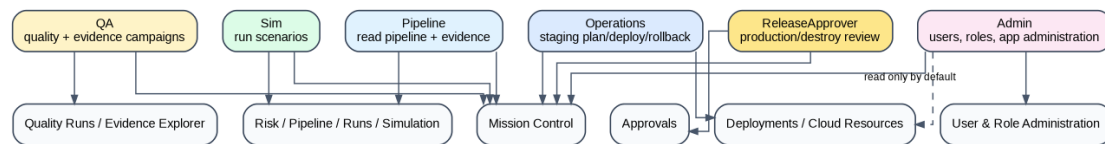
The browser remains outside the provider trust boundary. It selects a closed operation identifier; the backend authorises and records it; specialised runners hold the provider identity.



Operations lifecycle

Roles and separation of powers

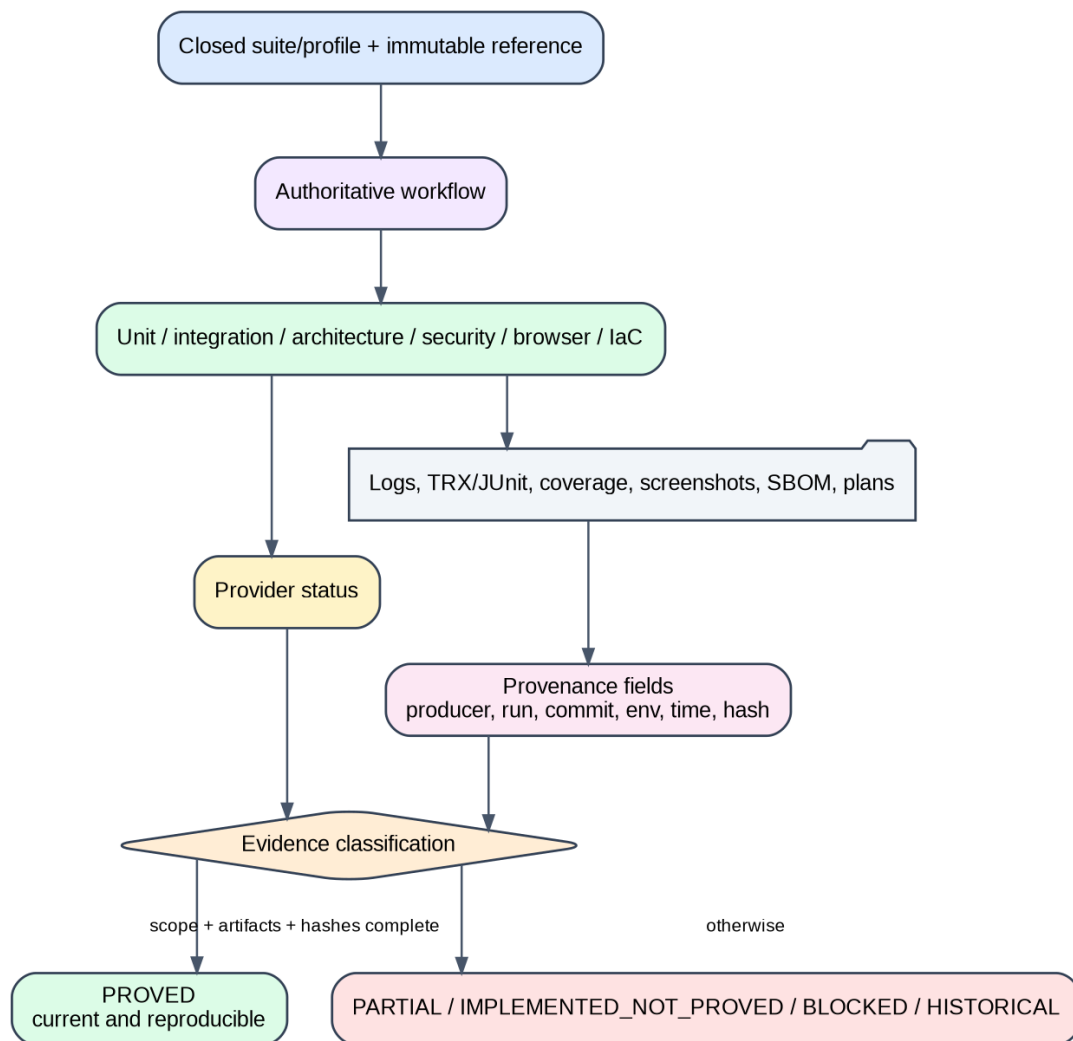
The current role model adds QA, Operations and ReleaseApprover. Admin manages application users and roles but does not automatically receive production deploy or destroy capability. The distinction remains meaningful even when one academic project owner holds several roles, because confirmation and approval steps remain separately recorded.



Roles and UI journeys

Quality and evidence

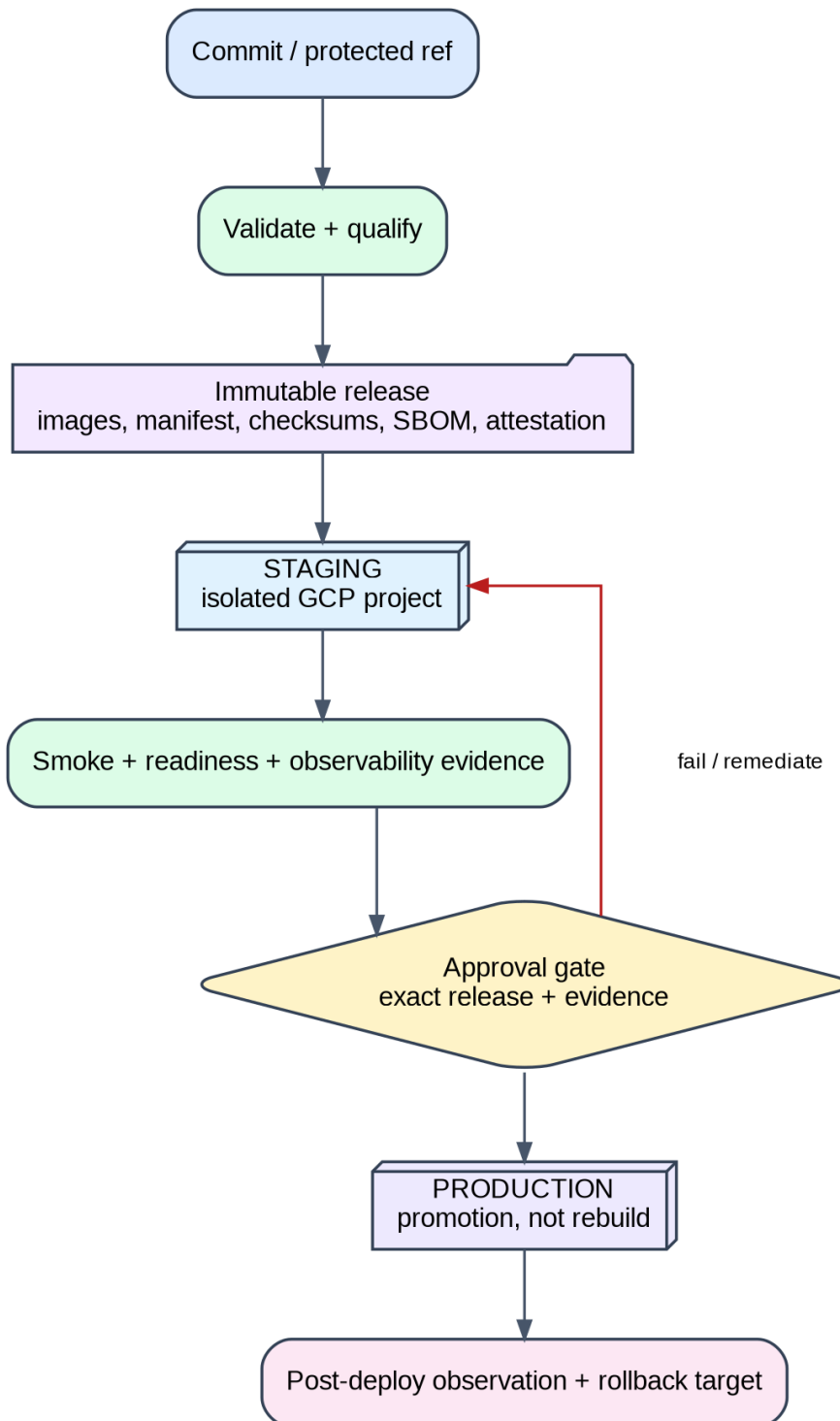
Quality and evidence operations are selected from a bounded catalog and dispatched to authoritative workflows. A provider success status is not enough for a **PROVED** claim. The expected artifacts, snapshot reference, environment, producer and SHA-256 must be present and the scope must match the claim.



Quality and evidence flow

Cloud and deployment

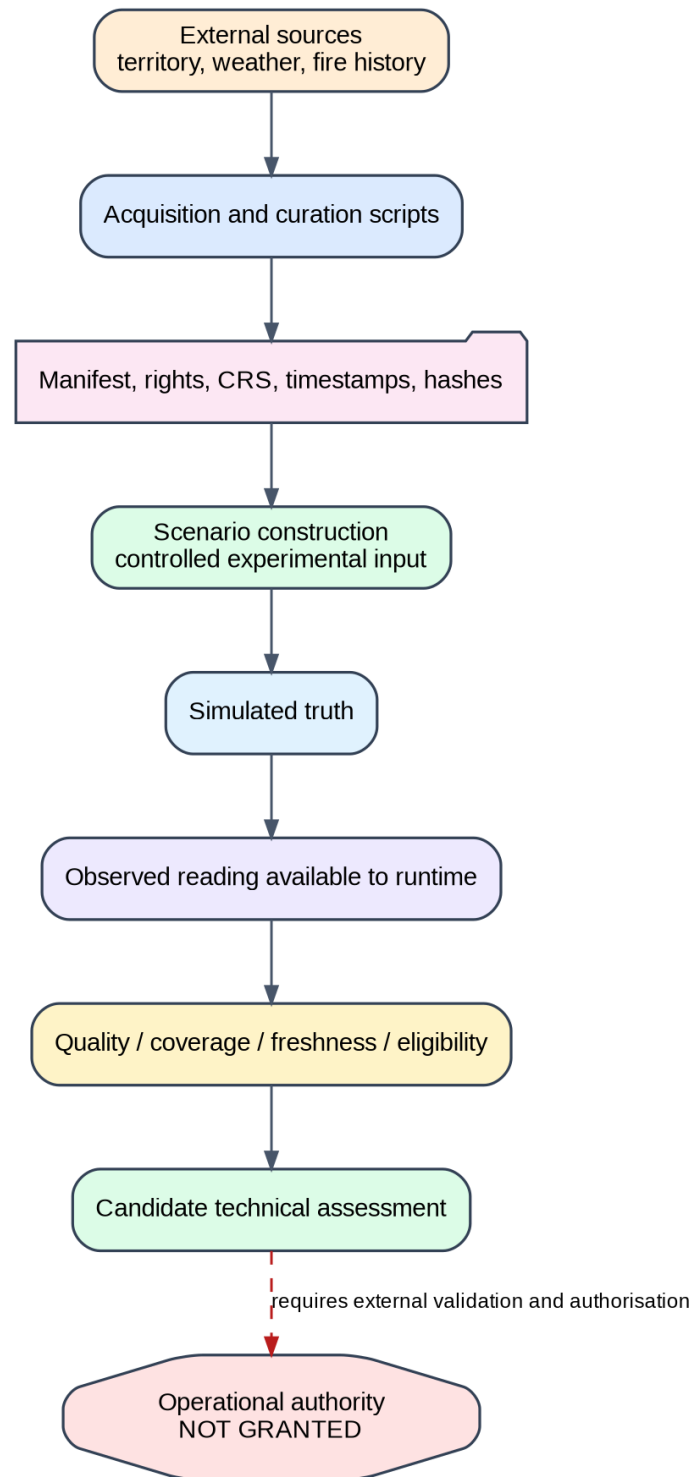
The implementation contains Terraform, GKE/Autopilot, Cloud Deploy, Artifact Registry, WIF and release/deployment workflows. The correct factual statement remains that the source implementation is advanced but the supplied artifacts do not prove a signed release for the final head, a qualified staging deployment or production.



Deployment and promotion

Scientific and operational boundary

The current engineering progress does not change the report's scientific limits. Candidate FWI, KBDI and NatureProtector calculations remain technical comparison components rather than official, calibrated or validated operational methods.



Data provenance and authority

Chapter integration map

Target	Integration
Chapter 4	Add operations plane, runner/provider boundary and environment views
Chapter 5	Add the unified convergence loop and documentation governance
Chapter 6	Add capabilities, operation catalog, pages, workflows and callback
Chapter 8	Add closed quality/evidence dispatch and hashed-artifact classification
Chapter 9	Discuss separation of administrative and release authority
Chapter 10	Remove already-implemented Operations UI from future work; retain runtime/cloud/scientific proof
Chapter 11	Extend auditability contribution without claiming production proof
Appendix E	Update components, endpoints, roles and repository structure
Appendix H	Update quality/evidence tooling and open execution gates
Appendix I	Update administration and capabilities
Appendix J	Update cloud/CD implementation and proof boundary
Appendix K	Use promoted current diagrams and current UI captures when available

Wording controls

Use: implemented, statically verified, identified execution, historical evidence, current snapshot, partial, blocked, candidate, experimental.

Do not use without current proof: production-ready, live, real-time, validated, calibrated, official alert, operational system, real data.

Remaining integration requirement

The editable source of the 379-page current report was not included. A future editorial operation must insert the supplied LaTeX addendum and chapter-level deltas into that source, recompile the report and perform a full visual and claim-traceability review.